

Bogoan Kim, Ph.D.

CONTACT INFORMATION	Building E8-10, Room 302 Chungdae-ro 1, Seowon-gu, Cheongju-si, Chungcheongbuk-do, 28644, Korea	Tel: + 82 - 43 - 261 - 2483 Email: bogoan@cbnu.ac.kr Homepage: https://bogoankim.github.io/
RESEARCH INTEREST	My research sits at the intersection of human-computer interaction (HCI) , digital health , and inclusive AI . I design and empirically evaluate AI-augmented systems—spanning mobile applications, immersive (VR) environments, and LLM-powered tools—grounded in multimodal behavioral, physiological, and contextual data. A particular emphasis is placed on preserving human agency in AI-driven processes and on the meaningful integration of AI into real-world workflows across diverse professional and social contexts, informed by participatory design and evidence synthesis.	
POSITIONS	Chungbuk National University , Cheongju, Korea • <i>Assistant Professor, School of Information and Communication Engineering</i> • <i>Affiliated Researcher, Cancer Research Center</i> • <i>Director, CBNU HCI Lab</i> KAIST , Daejeon, Korea • <i>Jang Young-Sil Postdoctoral Fellow, College of Engineering</i> – Host Professors: Profs. Hwajung Hong and Uichin Lee The Pennsylvania State University , PA, USA • <i>Visiting Scholar, College of IST</i> – Mentors: Profs. Dongwon Lee and Aiping Xiong Indiana University , IN, USA • <i>Visiting Scholar, Luddy School of Informatics</i> – Mentors: Profs. Patrick C. Shih and Georgia C. Frey	Sep 2025 – Present

IITP High-Potential Individuals Global Training Fellowship	2020
• Support for research @ Pennsylvania State University	
Indiana University Global Talent Attraction Program (GTAP) Fellowship	2019
• Support for research @ Indiana University	
Honors on the Dean's List	2011, 2012, 2015
Global Excellence Scholarship (merit-based; 4,210 USD)	2011
Special Recognitions for Outstanding Reviews	
• CHI 2024, CHI 2025, IMWUT 2025, CHI 2026	
Domestic Conference Awards	
• Grand Paper Award: KDMS 2024	
• Best Paper Award: KDMS 2024	
• Best Presentation Award: KCC 2024	

PUBLICATIONS

Conference Proceedings (★: [NRF top-tier CS conferences](#); *: equal contribution)

[c.13] When Special Education Meets LLMs: Investigating the use of LLM-based Conversational AI by Special Education Teachers for Autism in China

J Wu, D Choi, [B Kim](#), and H Hong

DIS 2026 (*ACM International Conference on Designing Interactive Systems*)

Best Paper Honorable Mention Award (Top 5%)

[c.12] QuerySwitch: Supporting the Design Process by Balancing Vagueness through Large Language Models

★ M Kim, [B Kim](#), and K Han

CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#) [10-min Talk](#) [Slides](#) [Media](#)

[c.11] LAPS: Automating Hypothesis-Driven Statistical Analysis of Public Survey Using Large Language Models

★ {J Kim*, D Jeong*}, B Son, H Kim, [B Kim](#), and K Han

CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#) [10-min Talk](#) [Slides](#) [Media](#)

Best Paper Honorable Mention Award (Top 5%)

[c.10] “I Choose to Live, for Life Itself”: Understanding Agency of Home-Based Care Patients Through

★ Information Practices and Relational Dynamics in Care Networks

{S-I Kim*, J Park*}, [B Kim](#), and H Hong

CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#)

[c.9] Understanding Human-Multi-Agent Team Formation for Creative Work

★ H Lim, D Choi, S Nam, [B Kim](#), and H Hong

CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#)

[c.8] “I Don’t Know Why I Should Use This App”: Holistic Analysis on Engagement Challenges in Mental Health

★ {S Jin*, [B Kim*](#)}, and K Han

CHI 2025 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#) [Media](#)

[c.7] Narrating Routines through Game Dynamics: Impact of a Routine Management App for Autistic Individuals

★ [B Kim](#), D Jeong, H Hong, and K Han

CHI 2024 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#) [10-min Talk](#) [System Teaser](#)

[c.6] V-DAT: Supporting Self-Awareness for Autistic People from Multimodal VR Sensor Data

★ [B Kim](#), D Jeong, JG Kim, H Hong, and K Han

UIST 2023 (*ACM Symposium on User Interface Software and Technology*)

Links : [PDF](#) [10-min Talk](#) [System Teaser](#)

- [c.5] Supporting Independence of Autistic Adults through Mobile and Virtual Reality Technologies
 * B Kim (Doctoral Symposium)
UIST 2023 (*ACM Symposium on User Interface Software and Technology*)
 Doctoral Symposium
 Links : [PDF](#)
- [c.4] RoutineAid: Externalizing Key Design Elements to Support Daily Routines of Individuals with Autism
 * B Kim, S-I Kim, S Park, HJ Yoo, H Hong, and K Han
CHI 2023 (*ACM CHI Conference on Human Factors in Computing Systems*)
 Links : [PDF](#) [10-min Talk](#) [System Teaser](#)
- [c.3] VISTA: User-Centered VR Training System for Effectively Deriving Characteristics of Autistic Individuals
 * B Kim, D Jeong, M Jeong, T Noh, S-I Kim, T Kim, S-Y Jang, HJ Yoo, JG Kim, H Hong, and K Han
VRST 2022 (*The ACM Symposium on Virtual Reality Software and Technology*)
 Links : [PDF](#) [10-min Talk](#) [System Teaser](#) [Media](#)
- [c.2] ChamberBreaker: Mitigating the Echo Chamber Effect through a Gamified Inoculation System
 * Y Jeon, B Kim, A Xiong, D Lee, and K Han
CSCW 2021 (*ACM Conference on Computer-Supported Cooperative Work and Social Computing*)
 Links : [PDF](#) [Media](#)
- [c.1] Photos Don't Have Me, But How Do You Know Me?: Analyzing and Predicting Users on Instagram
 K Han, Y Jo, Y Jeon, B Kim, J Song, and S-W Kim
UMAP 2018 (*ACM Conference on User Modeling, Adaptation and Personalization*)
 Links : [PDF](#)

Journal Articles (*: equal contribution)

- [j.11] VR-ACT: Transforming Clinical Evidence of ACT through VR for Subthreshold Depression
B Kim, D Jeong, YS Choi, H Kim, and K Han
Springer Virtual Reality (*SCIE; IF 5.0; JCR 11.2%; Published 2026.03*)
 Links : [PDF](#)
- [j.10] Mobile and VR Technologies for Social Independence in Autistic Individuals: A Scoping Review
B Kim, D Kang
Journal of Korean Institute of Intelligent Systems (*KCI; Published 2026.03*)
- [j.9] Enhancing Mindfulness-Based Cognitive Therapy in a Virtual Reality: A Prospective Interventional Study
 {B Kim*, D Jeong*}, YS Choi, Y Choi, H Kim, and K Han
Scientific Reports (*SCIE; IF 3.9; JCR 18.0%; Published 2025.09*)
 Links : [PDF](#)
- [j.8] Development of a Personalized Early Prediction Model for Cybersickness in Virtual Reality
 YS Choi, D Jeong, B Kim, and K Han
KIISE Transactions on Computing Practices (*KCI; Published 2025.06*)
- [j.7] Promoting Self-Efficacy of Individuals with Autism in Practicing Social Skills in the Workplace Using Virtual Reality and Physiological Sensors: Mixed Methods Study
 S-I Kim, S-Y Jang, T Kim, B Kim, D Jeong, T Noh, M Jeong, HJ Yoo, K Han, H Hong, and JG Kim
JMIR Formative Research (*SCIE; IF 2.20, Published 2024.01*)
 Links : [PDF](#) [Video](#)
- [j.6] A Study on Mobile Game Design to Support Daily Routines of Individuals with Autism
B Kim, S-I Kim, S Park, HJ Yoo, H Hong, and K Han
Journal of the HCI Society of Korea (*KCI; Published 2022.05*)
- [j.5] Usability Assessment of a Mobile Health Intervention for Increasing Physical Activity in Autistic Adults
 DH Lee, GC Frey, A Min, B Kim, DJ Cothran, S Bellini, K Han, and PC Shih
Health Informatics Journal (*SCIE; IF 2.3; Published 2020.09*)
 Links : [PDF](#)

- [j.4] PuzzleWalk: A Theory-Driven Iterative Design of a Mobile Game for Promoting Physical Activity in Autistic Adults
B Kim, DH Lee, A Min, S Paik, GC Frey, S Bellini, K Han, and PC Shih
PLoS ONE (SCIE; IF 2.9; Published 2020.09)
Links : [PDF](#)
- [j.3] No More One Liners: Bringing Context into Emoji Recommendations
J-G Kim, T Gong, B Kim, J Park, W Kim, E Huang, K Han, J Kim, JG Ko, and S-J Lee
ACM Transactions on Social Computing 2020
Links : [PDF](#)
- [j.2] A Systematic Review on Fake News Research through the Lens of News Creation and Consumption: Research Efforts, Challenges, and Future Directions
B Kim, A Xiong, D Lee, and K Han
PLoS ONE (SCIE, IF = 2.9, Published 2020.09)
Links : [PDF](#)
- [j.1] ATM: The Effectiveness of Self-Regulation on Time Management through a Smartphone Application
B Kim, S-W Lee, H Hong, and K Han
IEEE Access (SCIE, IF = 3.4, Published 2019.07)
Links : [PDF](#)

Posters and Workshops

- [p.9] Agreement Between Smartphone- and Research-grade Accelerometer-measured Step Counts in Autistic Adults in Free-living Conditions
L McNulty, B Kim, AE Bodde, ML Mauriello, BC Helsel, and D Lee
ACSM 2026 (*American College of Sports Medicine*)
- [p.8] Early Prediction of Cybersickness in Virtual Reality Using an LLM for Multimodal Time Series Data
YS Choi, D Jeong, B Kim, and K Han
UbiComp 2024 (*ACM Conference on Interactive, Mobile, Wearable and Ubiquitous Technologies*)
- [p.7] Objectively Measured Physical Activity and Sedentary Time in Autistic Adults
D Lee, PC Shih, B Kim, K Han, A Min, D Cothran, S Bellini, and GC Frey
ACSM 2020 (*American College of Sports Medicine*)
- [p.6] A Study on the Text-Based Characteristics of Autistic Individuals through Social Media Data Analysis
J Han, B Kim, and K Han
ICNGC 2020 (*The International Conference on Next Generation Computing*)
- [p.5] FashionQ: An Interactive Tool for Analyzing Fashion Style Trend with Quantitative Criteria
Y Jeon, S Jin, B Kim, and K Han
CHI 2020 (*ACM Conference on Human Factors in Computing Systems*)
- [p.4] Ecological Momentary Assessment of Physical Activity and Anxiety in Autistic Adults
D Lee, D Cothran, S Bellini, PC Shih, K Han, B Kim, A Min, S Paik, and GC Frey
INSAR 2020 (*International Society for Autism Research*)
- [p.3] Bringing Context into Emoji Recommendations
J-G Kim, T Gong, B Kim, J Park, E Huang, W Kim, K Han, J Kim, JG Ko, and S-J Lee
MobiSys 2019 (*ACM Conference on Mobile Systems, Applications, and Services*)
- [p.2] Iterative Design and Usability of Gamified Application for Adults with Autism Spectrum Disorder
S Paik, B Kim, and K Han
ICNGC 2019 (*The International Conference on Next Generation Computing*)
- [p.1] Leveraging Mobile Technology to Support Goal Setting and Strategies of College Students
B Kim, S Rhim, K Han, and S-W Lee
UbiComp 2018 (*ACM Conference on Interactive, Mobile, Wearable and Ubiquitous Technologies*)

PATENTS & SOFTWARE

Granted Patents

- VR-based Interactive Social Skills Training System for Autistic Individuals
Registration Number: **KR10-2938949** Mar 2026
- A Mobile System for Routine Management in Autistic Individuals
Registration Number: **KR10-2729263** Oct 2024

Filed Patents

- A Multimodal LLM Capturing Inter-modal Dynamics for Personalized VR Sickness Prediction
Application Number: **KR10-2025-0100532** Jun 2025
- VR Data Analysis Tool to Support Self-Awareness for Autistic Individuals
Application Number: **KR10-2023-0116955** Aug 2023

Registered Software

- VR-based Dashboard for Clinical Support and Patient Monitoring in Depression Care
Registration Number: **C-2024-034698** Oct 2024

TEACHING EXPERIENCE

Faculty Instructor (*: courses taught in English)

- [GR] Special Topics in Deep Learning* (8836552-01) Spring 2026
- [UG] Information Technology Programming (0914006-10) Spring 2026
- [UG] Gaesin Frontier II (0281087-12) Spring 2026
- [UG] Capstone Design (5108102-03) Spring 2026
- [UG] Advanced Computer Programming (5108113-01) Spring 2026
- [UG] Intelligent System (5108115-01) Fall 2025
- [UG] Object-Oriented Programming (5108124-01) Fall 2025
- [UG] Fundamental Computer Programming (0914002-18) Fall 2025

Instructor / Co-lecturer (*: courses taught in English)

- Computational Social Science* Spring 2025
- Human-Computer Interaction* Fall 2024
- Human-Computer Interaction* Fall 2023
- Introduction to Software Design* Spring 2023
- Creative Software Design* Fall 2022
- Introduction to Software Design* Spring 2022
- Creative Software Design* Fall 2021
- Data Structures and Practice* Spring 2021
- AI Intensive Class Fall 2019
- Data Structures and Practice* Spring 2019

Teaching Assistant

- Algorithms Fall 2020
- Introduction to AI Programming Language Spring 2020
- The Basics of Data Analytics Fall 2018
- The Basics of Data Analytics Spring 2018

INVITED TALKS

Conference Invited Talk, Dajeon, Korea

- Host: Korean Society of Veterinary Science Spring Conference 2026 Apr 2026
- Session: The Next-Generation Research Paradigm in the Era of AI Transformation (AX)
- Topic: Designing Agency-Centered Interactions in Digital Health

Lab Guest Speaker Session, Seoul, Korea

- Host: MINDS Lab @ Chung-Ang University Aug 2025
- Topic: Designing Sociotechnical Systems for People in Need

Europe-Korea Conference 2025 Seminar, Seoul, Korea

- Host: College of Electrical & Computer Engineering @ Chungbuk National University Aug 2025
- Topic: The Role of HCI Researchers in the Development of Digital Therapeutics

Lab Guest Speaker Session , Daejeon, Korea	May 2025
• Host: DxD Lab @ KAIST • Topic: AI-Infused Software Systems for Solving Real-World Challenges	
Omnibus Lecture Series on Computational Social Science , Seoul, Korea	Mar 2025
• Host: Department of Data Science @ Hanyang University • Topic: User-Centered Digital Healthcare	
Computational Social Science Colloquium , Seoul, Korea	May 2024
• Host: Center for Computational Social Science @ Hanyang University • Topic: Sociotechnical Systems for People Who Are in Need	
Top Conference Session Invited Talk , Hongcheon, Korea	Jan 2024
• Host: HCI Korea 2024 • Topic: Impact of a Gamified Routine Management App for Autistic Adults	
UIST Doctoral Symposium , San Francisco, CA USA	Oct 2023
• Host: ACM UIST 2023 • Topic: AI-VR Framework for Personalized Adaptive System	
Top Conference Session Invited Talk , Jeju, Korea	Jun 2023
• Host: Korea Computer Congress (KCC) 2023 • Topic: Externalizing Key Design Elements to Support Daily Routines of Autistic Adults	
IITP Global Talent Program Workshop , Seoul, Korea	Dec 2021
• Host: Institute of Information & Communications Technology Planning & Evaluation (IITP) • Topic: Explainable Fake News Detection with Deep Learning and Human Factors	
HCI Korea Doctoral Colloquium , Seoul, Korea	Jan 2021
• Host: HCI Korea 2021 • Topic: Designing Inclusive Mobile Applications for Marginalized Populations	
Lab Guest Speaker Session , PA, USA	Feb 2020
• Host: The PIKE Group @ The Pennsylvania State University • Topic: From Creation to Consumption: Data-Driven Perspectives on Fake News Research	
Invited Guest Lecture , Suwon, Korea	Oct 2019
• Host: AI Intensive Class @ Ajou University • Topic: Designing a Gamified mHealth App for Marginalized Populations	
Global Talent Attraction Program Conference , IN, USA	Jun 2019
• Host: Luddy School of Informatics, Computing, and Engineering @ Indiana University • Topic: Designing a Mobile Game to Promote Physical Activity in Autistic Adults	
Global Challenge Program Symposium , Tokyo, Japan	Jun 2018
• Host: Environment and Information Studies @ Keio University • Topic: Effectiveness of a Smartphone App in Self-Regulating Time Management	

ACADEMIC SERVICES

Conference Reviewer	
• The Conference on Human Factors in Computing Systems (CHI) • The Conference on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT) • The Designing Interactive Systems Conference (DIS) • The Conference on Computer-Supported Cooperative Work & Social Computing (CSCW) • The Conference on Information and Knowledge Management (CIKM) • The Symposium on Applied Computing (SAC) • The Symposium on Mixed and Augmented Reality (ISMAR)	2019 - 2026 2019 - 2025 2025 2019 - 2024 2021 - 2023 2022 2021 - 2022
Journal Reviewer	
• npj Digital Medicine • ACM Transactions on Accessible Computing • JMIR Formative Research • The Nature Scientific Reports • The Taylor and Francis Interactive Learning Environments	2026 2026 2025 2024 2023

Student Volunteer

- The Conference on Mobile Systems, Applications, and Services (**MobiSys**)

2019

ACADEMIC MENTORSHIP

Mentored Students

- C Lee, Chungbuk National University (Master's Student) Fall 2025 - Present
- S Seo, Chungbuk National University (Undergrad. Student) Fall 2025 - Present
- D Jeong, Chungbuk National University (Undergrad. Student) Fall 2025 - Present
- D Lee, Chungbuk National University (Undergrad. Student) Fall 2025 - Present
- J Kim, Hanyang University (Ph.D. Student) Spring 2025 - Spring 2025
- L McNulty, University of Delaware (Ph.D. Student) Spring 2025 - Fall 2025
- H Lim, KAIST (Ph.D. Student) Spring 2025 - Fall 2025
- S-I Kim, Seoul National University (MD, Ph.D. Student) Fall 2024 - Fall 2025
- J Park, KAIST (Master's Student) Spring 2025 - Fall 2025
- J Wu, KAIST (Ph.D. Student) Spring 2025 - Fall 2025
- M Kim, Hanyang University (Ph.D. Student) Spring 2024 - Present
- YS Choi, Hanyang University (Master's Student) Fall 2023 - Present
- D Jeong, Hanyang University (Ph.D. Student) Spring 2022 - Present
- S Paik, Ajou University (Master's Student) Spring 2019 - Fall 2020

PRESS & INTERVIEWS

Research Media Coverage

- "충북대 김보관 교수 연구팀, 우수논문상 2편 수상" Apr 2026
- "인간의 가치를 지키는 'AI 가치 정렬' 연구 성과..." Mar 2026
- "정신건강관리 앱 지속사용 한계..." May 2025
- KAIST 등, 자폐스펙트럼장애 직장 적응 돕는 VR 개발 Jan 2023
- New game can help users identify, avoid online echo chambers Oct 2021

REFERENCES

Hwajung Hong, *Professor* (Postdoc. Host Professor)
Department of Industrial Design, KAIST

hwajung@kaist.ac.kr

Uichin Lee, *Endowed Chair Professor* (Postdoc. Host Professor)
School of Computing, KAIST

uclee@kaist.ac.kr

Kyungsik Han, *Professor* (Ph.D. Advisor)
Department of Data Science, Hanyang University

kyungsikhan@hanyang.ac.kr

Daehyoung Lee, *Assistant Professor* (Collaborator)
Department of Health Behavior and Nutrition Sciences, University of Delaware

dhlee@udel.edu

Jennifer G. Kim, *Assistant Professor* (Collaborator)
School of Interactive Computing, Georgia Institute of Technology

jennifer.kim@cc.gatech.edu

Dongwon Lee, *Professor* (Collaborator)
College of IST, The Pennsylvania State University – University Park

dongwon@psu.edu

Patrick C. Shih, *Associate Professor* (Collaborator)
Luddy School of Informatics, Indiana University – Bloomington

patshih@indiana.edu

Georgia C. Frey, *Associate Professor* (Collaborator)
School of Public Health, Indiana University – Bloomington

gfrey@indiana.edu

Aiping Xiong, *Associate Professor* (Collaborator)
College of IST, The Pennsylvania State University – University Park

axx29@psu.edu